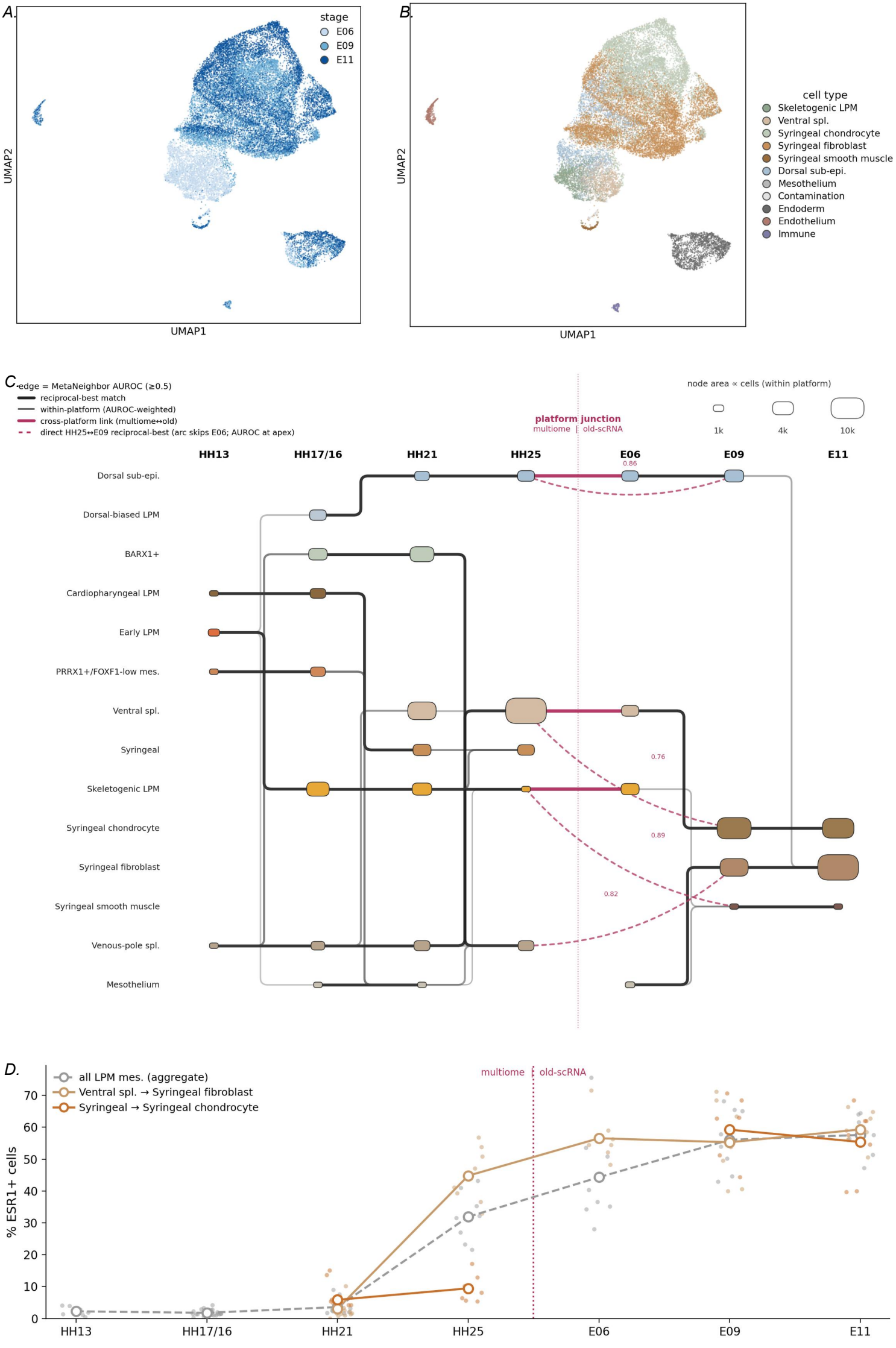
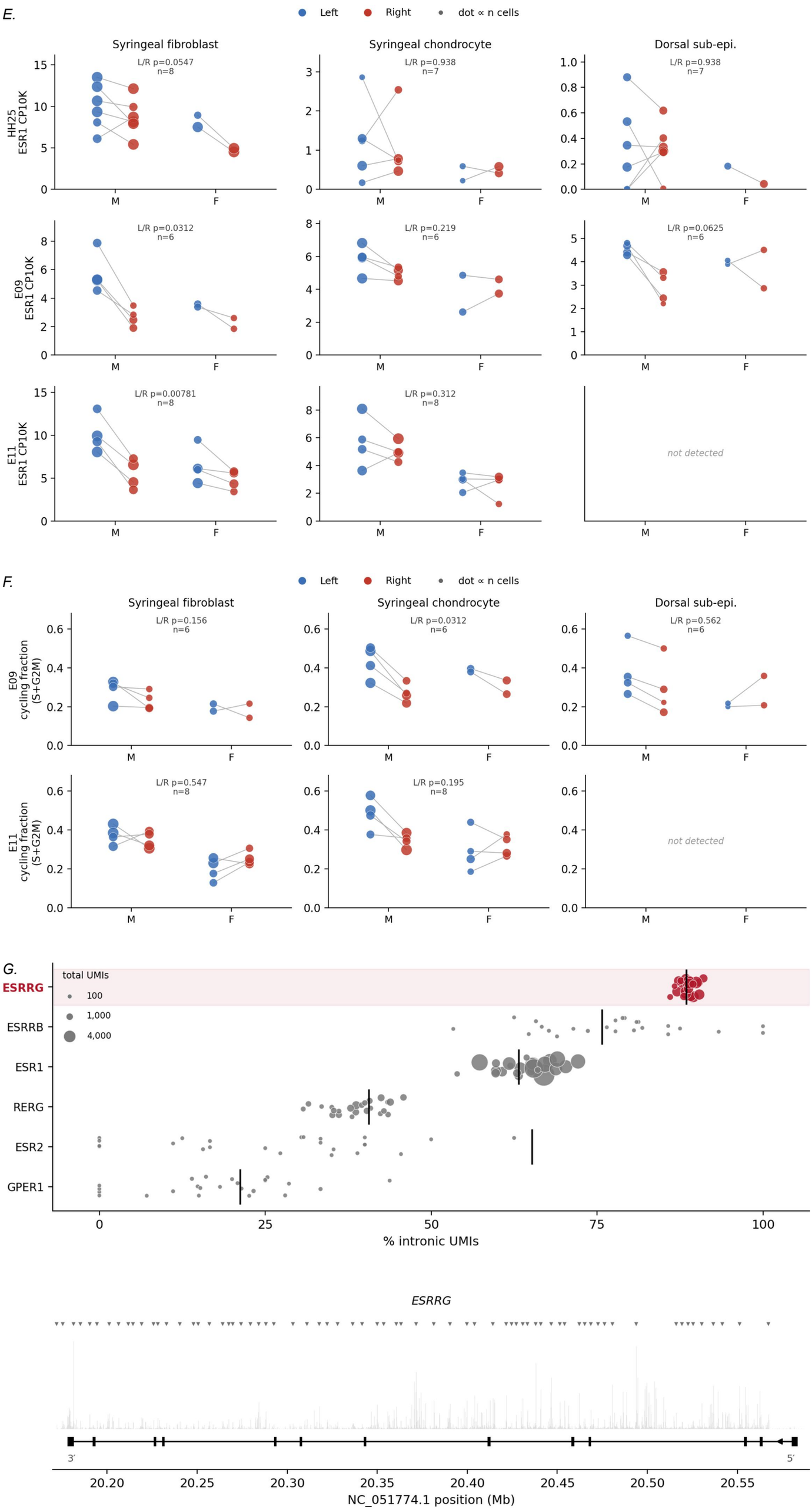

A.
stage
E06
E09
E11
UMAP2
UMAP1
B.
cell type
Skeletogenic LPM
Ventral spl.
Syringeal chondrocyte
Syringeal fibroblast
Syringeal smooth muscle
Dorsal sub-epi.
Mesothelium
Contamination
Endoderm
Endothelium
Immune
UMAP2
UMAP1
C.
edge = MetaNeighbor AUROC (≥0.5)
reciprocal-best match
within-platform (AUROC-weighted)
cross-platform link (multiome↔old)
direct HH25↔E09 reciprocal-best (arc skips E06; AUROC at apex)
node area ∝ cells (within platform)
1k
4k
10k
platform junction
multiome | old-scRNA
HH13
HH17/16
HH21
HH25
E06
E09
E11
0.86
0.76
0.89
0.82
Dorsal sub-epi.
Dorsal-biased LPM
BARX1+
Cardiopharyngeal LPM
Early LPM
PRRX1+/FOXF1-low mes.
Ventral spl.
Syringeal
Skeletogenic LPM
Syringeal chondrocyte
Syringeal fibroblast
Syringeal smooth muscle
Venous-pole spl.
Mesothelium
D.
all LPM mes. (aggregate)
Ventral spl. → Syringeal fibroblast
Syringeal → Syringeal chondrocyte
multiome | old-scRNA
% ESR1+ cells
0
10
20
30
40
50
60
70
HH13
HH17/16
HH21
HH25
E06
E09
E11


E.
Left
Right
dot ∝ n cells
Syringeal fibroblast
Syringeal chondrocyte
Dorsal sub-epi.
HH25 ESR1 CP10K
E09 ESR1 CP10K
E11 ESR1 CP10K
L/R p=0.0547 n=8
L/R p=0.938 n=7
L/R p=0.938 n=7
L/R p=0.0312 n=6
L/R p=0.219 n=6
L/R p=0.0625 n=6
L/R p=0.00781 n=8
L/R p=0.312 n=8
not detected
M
F
F.
E09 cycling fraction (S+G2M)
E11 cycling fraction (S+G2M)
L/R p=0.156 n=6
L/R p=0.0312 n=6
L/R p=0.562 n=6
L/R p=0.547 n=8
L/R p=0.195 n=8
G.
total UMIs
100
1,000
4,000
ESRRG
ESRRB
ESR1
RERG
ESR2
GPER1
% intronic UMIs
ESRRG
3′
5′
NC_051774.1 position (Mb)